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eva3dm: A R-package for model evaluation of 3D weather and air quality models

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Software

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Summary

Eva3dm is a package designed to support the evaluation of 3-dimensional physical models (in special, weather and air quality models) against observation data in order to quantify different errors and bias present in the model results and/or classify the model performance based on criteria from the literature.

Statement of need

Evaluation is the most important step for any model application, it can not only assure that the model results represent (or not) accurately the interest variable but also can guide to choose between different model configurations and guide model development. There are currently other tools available in R (Carslaw & Ropkins, 2012), Python (Ladwig, 2017) or other languages (Appel, 2011; NCAR et al., 2017) but none provide the degree of integrated support presented in this package: from the pre-processing of observations and model output, evaluation and visualization.

Description

In the literature there are many different evaluation criteria depending on the application and variable (Emery, 2001, 2017; Monk, 2019; Ramboll, 2018; Zhai, 2024; Zhang, 2019). These criteria varies depending on the goal of the simulation, the variability of the observations, the error associated to the measurements, and how other models available in the literature compared with observations and should be used accordingly to the particular application, study region and other uncertainties present on the model and model inputs.

The package was designed to support the pre-processing of observations, post-processing of model output, evaluation and visualization of the results from the evaluation. A brief description of these 4 steps is provided bellow:

1. **Pre-processing of observations**:
    - Download of the observations, example of observations available:
        - METAR of meteorological observations can be downloaded using the R-package riem or via the Iowa State University site
        - AERONET data for Aerosol Optical Depth can be downloaded at AErosol RObotic NETwork
        - Air Quality data for Brazil can be downloaded using the R-package qualR, or QUALAR and MonitorAir sites
        - Satellite products are available at NASA giovanni website

- Process a list of the location to extract time-series from the model. NOTE: There are examples of list of sites in the documentation of `extract_serie` for METAR, AERONET, INMET in Brazil, and Air Quality stations in Brazil.
- Process observation data for evaluation: Calculation of secondary variables (absolute humidity for example).
- Quality Assurance of the observation data: in this point the user should check for values that are outside the range of each variable (for example negative concentration and other potential problematic values), check if there is actual data from the time-period and region of the simulation and note any singular event that can be represented.

2. Pre-processing of model output: Extraction and post-processing of model outputs using `evad3m` functions and/or other tools and calculation of secondary variables (wind speed for example).

3. Model Evaluation: The evaluation consist in pair observations and model results and calculate statistical and/or categorical indexes that can be compared with the literature.

4. Visualization: There are functions to visualize model fields (taken into account different projections) and overlay of the statistical results.

To support the steps 1 to 4, the R-package includes the following functions:

Pre-processing of observations:

The functions `rh2q` and `q2rh` converts humidity units and the functions `mda8`, `ma8h`, `hourly`, and `daily` can be used to calculate time-series from observations and model. Most of the processing should be done by the user since the formats and conventions are different depending on the source of the data. The format used to evaluate time-series is a data.frame, the first column must contain time (in POSIXlt) and one additional column for each different location.

Pre-processing of model outputs:

The function `extract_serie` extract and save time-series from model outputs (or other compatible NetCDF files, more details in the documentation) using a data.frame with name (row names) and latitude (column lat) and longitude (column lon), while the functions `extract_mean` and `extract_max_8h` extract the average or the daily maximum of 8-hour moving average and save in a new NetCDF file;

The function `wrf_rast` can be used to read model output and return a `SpatRaster` or `SpatVector` object from the model files (and compatible NetCDF files) and its counterpart `rast_to_netcdf` that converts a `SpatRaster` to an array and/or save to a existing NetCDF file;

The functions `uv2ws` and `uv2wd` can be used to calculate wind speed and velocity from the model wind components (u and v) and the function `rain` can be used to calculate hourly precipitation from model accumulated precipitation variables.

Model evaluation functions:

There are two high level evaluation functions implemented in the package: `eva`, that does the temporal pairing of both model and observations by station (or combine all data) and the data of time-series and `sat` that interpolate and pair data in regular grids (it removes points from the boundary by default), these functions call the low level evaluation functions `stat` to calculate the statistical metrics or the `cate` to calculate categorical metrics in relation to a threshold value. These result can be written and read using the `write_stat` and `read_stat` functions.

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Visualization and extractting information functions:

There are functions for visualization, interpolation and to extract information from NetCDF files, Table 1 show a relation of the visualization functions.

Table 1: Visualization, interpolation and information functions.

Function name	Description
`plot_rast`	Custom plot for `SpatRaster` objects
`plot_diff`	Custom plot for absolute or relative difference of two `SpatRaster` objects
`overlay`	Custom plot to overlay points
`legend_range`	Custom legend that displays max, min and average
`interp`	Interpolation function that combines project and resample for `SpatRaster` objects
`ncdump`	Print a `ncdump -h` equivalent command for a NetCDF file
`vars`	Return the name of the variables on NetCDF file
`atr`	Read and write attributes from a Netcdf file

Figure 1 shows examples of the first 4 functions on Table 1:

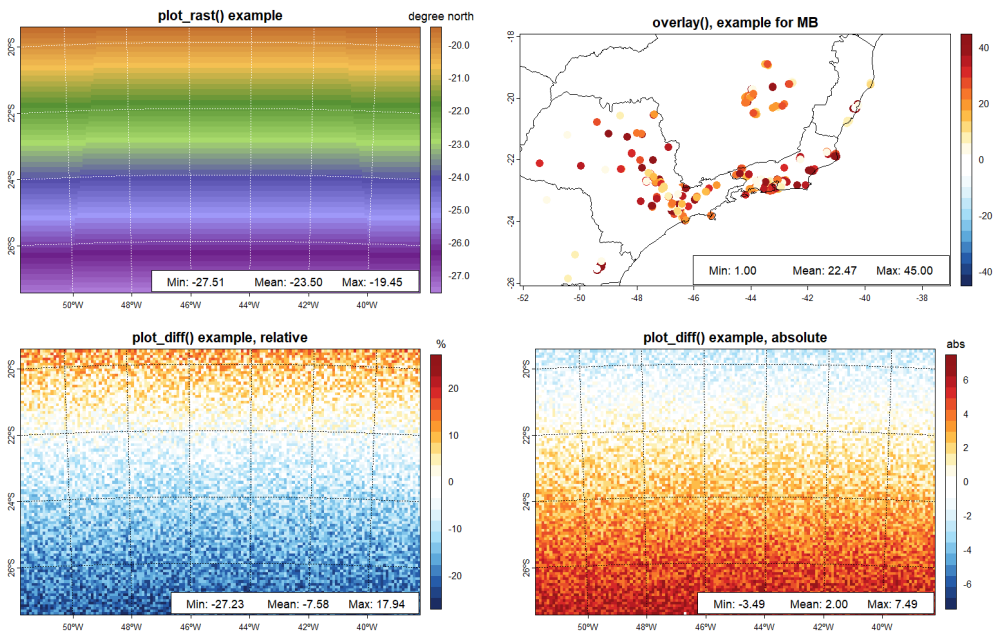

Figure 1: Figure 1 - Example from the viasualization functions.

Speciall functions:

There are tree special functions that make specific tasks:

Table 2: Special functions.

Function name	Description	Objective
`%at%`	Combine a data.frame containing evaluation results and a data.frame containing geographical coordinates (site list)	To georeference and visualize the statistical results combined with the `overlay` function
`%IN%`	Filter a data.frame (with time-series) based on a second data.frame. Also can be used to crop a `SpatRaster` based on a second `SpatRaster`	To perform fair comparison (using the same number of observation) of different simulations (with different domains for example) it can be combined with `eva` or `sat` functions
`template`	Create folders, post-processing and evaluation scripts. There are different templates that download observation data, process observations, post-process model output and evaluation for different groups of variables	To allow quickly process and evaluate multiple variables from one or multiple simulations (see the documentation for more details)

Most of the examples from `eva3dm` are focused on the Weather Research and Forecasting (WRF) model (Skamarock, 2024) and WRF coupled with Chemistry WRF-Chem (Grell, 2005), but other models, such as, the Comprehensive Air Quality Model with Extensions-CAMx (ENVIRON, 1998), Whole Atmosphere Community Climate Model-WACCM (Gettelman, 2019), and other models can be evaluated changing some of the arguments from the package functions from default.

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